Interleaving recovers the full circuit distance of bivariate-bicycle codes

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Strikis, Browne and Beverland showed that every non-interleaved single-ancilla syndrome-extraction circuit has circuit distance at most the distance d_{ext} of an extended code, which can fall below the code distance: at most 11 for the gross code $[[144, 12, 12]]$ and, as we show, at most 8 for $[[90, 8, 10]]$. We show that interleaving recovers the full code distance. We construct single-ancilla, interleaved circuits with circuit distance $d_{\mathrm{circ}} = 12$ for the gross code, answering their open question, and $d_{\mathrm{circ}} = 10$ for $[[90, 8, 10]]$, two above every non-interleaved circuit, and prove both under standard circuit-level depolarizing noise in both memory bases for every number of rounds. The method keeps a fixed extended code and turns circuit design into a finite decision problem over its low-weight logicals, which we solve exactly and certify with checked DRAT proofs. A direct search on the full detector error model confirms both results independently of the extended code, and our computation also confirms their conjecture that their original gross circuit has $d_{\mathrm{circ}} = 11$. The new circuits have periods of 18 and 16 time steps per round, against 8 for the IBM circuits.

Bivariate-bicycle (BB) codes [1] are leading candidates for low-overhead quantum memory; IBM's architecture is built on the gross code, with parameters $[[144, 12, 12]]$ [1, 2]. A code is only as good as the circuit that measures its checks: faults inside the circuit spread to several qubits, and the relevant figure of merit is the circuit distance d_{circ}, the minimum number of circuit faults that flip a logical observable without triggering any detector. At low physical error rates the logical error rate is governed by d_{circ}, not by the code distance d.

No circuit for the gross code had been shown to reach $d_{\mathrm{circ}} = d = 12$. The IBM circuit, with 8 time steps per round, has $d_{\mathrm{circ}} \leq 10$ [1, 2]. Strikis, Browne and Beverland (SBB) [3] analysed single-ancilla circuits, in which each check is measured by one dedicated ancilla coupled by CNOTs. They append to the check matrix the "residual" errors that an ancilla fault spreads onto the data, obtaining an extended code of distance d_{ext}, and show that $d_{\mathrm{circ}} \geq d_{\mathrm{ext}}$ for every such circuit (their Prop. 3), while $d_{\mathrm{circ}} \leq d_{\mathrm{ext}}$ holds when the circuit is *non-interleaved* in their sense, i.e. when all X-type CNOTs precede all overlapping Z-type CNOTs or vice versa (their Sec. II A and Cor. 1). Since every CNOT order gives $d_{\mathrm{ext}} \leq 11$ for the gross code, non-interleaved circuits are capped at 11 (their Prop. 4); their depth-8 circuit (their Fig. 12, hereafter "Fig. 12") was conjectured to reach it. They left open "whether any interleaved circuit can achieve circuit distance 12" [3]. For the $[[90, 8, 10]]$ code of the same family the best known value is 8: the IBM circuit has $d_{\mathrm{circ}} \leq 8$ [1], proven exact by MaxSAT [4].

We answer this question in the affirmative and show that the gap between Prop. 3 and Cor. 1 can be wider than one. For the gross code we construct a single-ancilla, CNOT-based, interleaved circuit (Table I) and prove $d_{\mathrm{circ}} = 12$ for every number of rounds (Theorem 1). For $[[90, 8, 10]]$ every choice of CNOT orders gives $d_{\mathrm{ext}} \leq 8$, so every non-interleaved circuit has $d_{\mathrm{circ}} \leq 8$; we construct an interleaved circuit (Table II) with $d_{\mathrm{circ}} = 10 = d$ (Theorem 2). The idea (Fig. 1) is to fix an extended code and choose CNOT orders and times such that none of its minimum-weight logicals, and none of the slightly heavier ones, can be realised by undetected circuit faults. This turns circuit design into a finite decision problem that we solve exactly and certify, a tool for single-ancilla circuits in general. Our computation of d_{ext} also proves SBB's conjecture that Fig. 12 has $d_{\mathrm{circ}} = 11$ (Corollary 1). The new circuits run with periods of 18 and 16 time steps per round instead of 8; we discuss this cost at the end.

Exact circuit distances are known for hypergraph-product codes [5], for the smallest BB circuit from exact solvers [6], and as certificates for other qLDPC constructions [7, 8]; for larger BB circuits reported values are typically upper bounds from randomized search. Shaw and Terhal's morphing circuit for the $[[288, 12, 18]]$ code uses the same number of qubits, has no dedicated ancillas, and at the end of each cycle acts on a $[[144, 12, 12]]$ code that they call "a second 'gross' code"; its circuit distance is reported as an upper bound, ≤ 12 [9]. Ref. [10] (App. C) describes a morphing idling circuit for the gross code that is "believed to have circuit distance 12", without giving the circuit; the circuit referred to there may coincide with the Shaw–Terhal construction. We note that this end-cycle code is not equivalent to the gross code under qubit permutations and local Cliffords, since its minimum stabiliser weight is 9 rather than 6. For $[[90, 8, 10]]$, Bi, Chang and Puri report that the upper bound for the unchanged IBM circuit returns to 10 when the ancilla noise is infinitely Z-biased [11]; we keep unbiased depolarizing noise and change the circuit instead, and our value is proven rather than a heuristic upper bound. SBB also show that the distance of an interleaved circuit is lower-bounded by that of its non-interleaved counterpart [3]; Ref. [12] proposed using this bound to screen interleaved schedules, leaving the search to future work. For the gross code that bound gives at most 11, so our route through a fixed extended code is different.

Check-dependent CNOT orders, by check type [13] or by colour class [3, 14], are an established tool.

Setting. We consider memory experiments in the Z or X basis: data initialisation, R rounds of syndrome extraction and transversal readout. Each check has one ancilla that is reset, coupled by six CNOTs and measured once per round; rounds of different checks may overlap in time (pipelining). Detectors compare consecutive measurements of a check, plus the first round and the final readout. Faults are the single-operation Pauli components of standard circuit-level depolarizing noise (SD6 [15]: two-qubit depolarization after CNOTs, flips after resets and before measurements, single-qubit depolarization on idle qubits). Faults with identical detector and observable effects are merged into one detector-error-model (DEM) mechanism [16], and d_{circ} counts distinct mechanisms: it is the minimum number of DEM mechanisms that flip a logical observable while triggering no detector. Lower bounds hold for any sub-model (e.g. without idle noise).

For a check with CNOT order $(q_1, \ldots, q_6)$, an ancilla fault after the k-th CNOT deposits the residual $\{q_{k+1}, \ldots, q_6\}$ on the data. The extended code of a basis appends all residual columns (Hs, Ls) of the relevant checks to the data columns (H: check matrix, L: logical operators). Every single fault, propagated to the readout, has a data image that is zero or one extended-code column (SBB, Table I and the proof of their Prop. 3; proved for our template in SM S5). For Fig. 12, $d_{\mathrm{ext}} = 11$ in both bases, with exactly 4224 weight-11 extended logicals per basis, all realisable in Fig. 12 by eleven undetected faults.

Key idea. We explain the idea on the gross code. Interleaving decouples d_{circ} from d_{ext} in one direction only: $d_{\mathrm{circ}} \geq d_{\mathrm{ext}}$ still holds, but d_{circ} can exceed it. To use this, the extended code must stay at distance 11, and the timing must prevent every weight-11 extended logical from lifting. We say that an extended logical *lifts* if one can choose, for each of its columns, a circuit fault whose propagated image is that column, such that the chosen faults together trigger no detector; a lift is then an undetected logical error of the same weight. The first requirement is met by an invariance. Up to the check itself, a residual of length 5 equals $\{q_1\}$, of length 4 equals $\{q_1, q_2\}$, of length 3 equals the complement of $\{q_1, q_2, q_3\}$, and of length 2 is $\{q_5, q_6\}$. The extended code therefore depends only on the three unordered sets $\{q_1, q_2\}$, $\{q_1, q_2, q_3\}$, $\{q_5, q_6\}$ of each check, which are preserved by reversal (rev) and by the swaps s_{12}, s_{56} of positions 1,2 and 5,6. All eight orders generated by these operations share the extended code of Fig. 12, its distance 11 and its list of 4224 weight-11 logicals (Fig. 1a).

The second requirement is a statement about time. Reversing an order keeps the extended code but changes which physical fault realises each residual column and when its spread reaches the neighbouring checks. Consider a weight-11 extended logical made of ten data errors and one two-qubit hook column $\{q_5, q_6\}$ of an X check (Fig. 1b). In Fig. 12 the column is produced by an ancilla fault after the fourth CNOT, which lands on the data after the neighbouring Z checks have read those qubits; every data error can then be placed after its reads, and no detector fires. With the order reversed, the same column can only come from a fault between the second and third CNOT, whose spread over four qubits (equal to the hook up to a stabiliser) reaches one neighbouring check after its readout and another before it. The checks along a cycle through the hook and three data qubits each see two errors of the logical and stay silent only if both fall on the same side of their readout; the early and late constraints propagate around the cycle and leave one check with odd parity, so a detector fires: timing turns the potential weight-11 logical into a parity cycle that cannot be satisfied. For all 360 logicals of this "ten data plus one hook" form, the lift fails at $R = 1$, and every irreducible conflicting set of detectors (one whose constraints alone forbid the lift, but no proper subset does) has between 4 and 10 detectors; the smallest such set has 4, 5 or 9 detectors. Circuit design becomes a finite problem: find orders in the class and CNOT times for which none of the 4224 known logicals lifts. With the Fig. 12 orders themselves it is impossible: every retiming family of Table S3 is obstructed (SM S11).

Construction. We searched schedules that are symmetric under the 24 translations preserving the colouring, choosing for each of the six (type, colour) classes one of the eight orders and its CNOT times with a SAT solver; constraints enforce legality (no data qubit in two CNOTs at once), determinism of all detectors, and blocking of every weight-11 extended logical in a bulk Pauli-frame model (SM S12). Candidates were then certified exactly. Table I and Fig. 2 give the resulting circuit. It measures the gross code itself, is legal and deterministic, is pipelined (every check's window crosses a period boundary) and interleaved (96 data qubits switch between X- and Z-type CNOTs four times per period, 48 twice); every check's order differs from that of Fig. 12. A second period-18 circuit and a period-21 circuit in which all ancillas are reset and measured together have the same distance (SM S2). For $[[90, 8, 10]]$ a search of the same kind over fully translation-invariant schedules, blocking the weight-8 and weight-9 extended logicals of Lemma 2 below, gave the circuit of Table II: all X checks share one order and timing, all Z checks another, every data qubit alternates X- and Z-type CNOTs six times per period, and the circuit is legal and deterministic.

Theorem 1 *Under SD6 noise, the circuit of Table I has* $d_{\mathrm{circ}} = 12$ *in both memory bases for every number of rounds* $R \geq 1$.

The proof rests on two lemmas that hold for every circuit of our template, and on one computation.

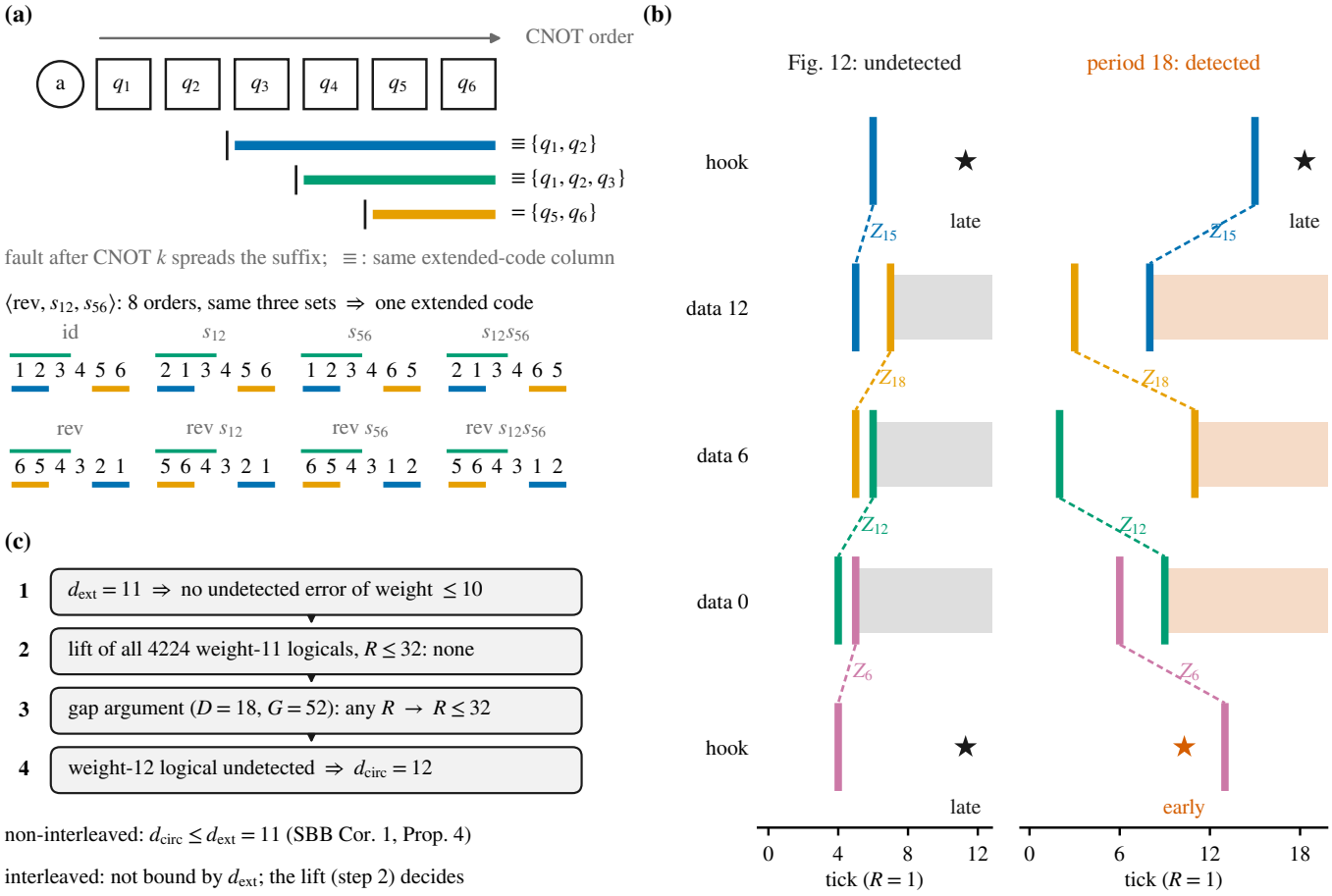


FIG. 1. The idea. (a) An ancilla fault after the k-th CNOT of a check spreads the residual $\{q_{k+1}, \ldots, q_6\}$ onto the data. Up to the check itself, the residuals of length 4, 3 and 2 give the same extended-code columns as $\{q_1, q_2\}$, $\{q_1, q_2, q_3\}$ and $\{q_5, q_6\}$, and those of length 5 and 1 act as single data qubits. The eight orders generated by reversal and the swaps s_{12}, s_{56} (top row: id, s_{12}, s_{56}, $s_{12}s_{56}$; bottom row: the same after reversal, e.g. "rev s_{12}" = reverse, then swap) keep these three sets and therefore share one extended code. (b) One weight-11 extended logical in a Z-basis memory with $R = 1$: ten data errors and the hook column $\{9, 25\}$ of an X check. Rows: the hook, data qubits 12, 6 and 0, and the hook again, closing the cycle; bars: the reads of the Z checks Z_{15}, Z_{18}, Z_{12}, Z_6, joined pairwise for each check; stars: when the hook error reaches the data, labelled late or early. Each of these checks sees exactly two errors of the logical and is silent only if both lie on the same side of its read. Left, Fig. 12: the hook lands on qubit 9 at tick 11, after its reads at ticks 6 and 4, and all data errors can be placed after their reads (grey); no detector fires. Right, period-18 circuit: the order of this check is reversed, so the residual after its fourth CNOT is no longer $\{9, 25\}$; the same extended-code column is realised only by an ancilla fault between the second and third CNOT, which spreads to the four qubits $\{8, 91, 82, 85\}$, equal to $\{9, 25\}$ up to the check itself. This spread is seen by Z_{15} on qubit 85 at tick 18, after its read at 15 (late), and by Z_6 on qubit 82 at tick 10, before its read at 13 (early). Being late for Z_{15} forces the data errors late for Z_{18}, Z_{12} and Z_6 (red), which leaves odd parity at Z_6, so a detector fires. (c) Logic of the proof: $d_{\text{ext}} = 11$ excludes weight ≤ 10 (Lemma 1); none of the 4224 weight-11 extended logicals lifts for $R \leq 32$ (Lemma 3); the gap argument reduces every R to $R \leq 32$ (Lemma 4); a weight-12 witness gives the upper bound. Non-interleaved circuits obey $d_{\text{circ}} \leq d_{\text{ext}} = 11$; for interleaved circuits the lift decides.

Lemma 1 (Reduction) *In any circuit of our template, an undetected fault set of size w that flips a logical observable yields an extended-code logical of weight at most w, formed by its nonzero fault images.*

Proof. Add one observable per row of H and L, evaluated on the final data readout. The observable of row H_i equals the sum of all detectors of check i, since these telescope from the first measurement to the final readout. For an undetected fault set the H parts of the images therefore sum to zero, while the L parts sum to the logical effect. Each image is zero or one column, a property noted by SBB (their Table I) that we prove for our template, interleaved and pipelined, in SM S5; pairs of equal columns cancel. □ This uses neither Prop. 3 of SBB nor non-interleaving; it excludes every weight below d_{ext}, for every R.

Lemma 2 (Light logicals) *An undetected logical of weight d_{ext} consists of d_{ext} faults whose images are dis-*

TABLE I. The period-18 circuit. Checks are grouped by type and colour $(x+y)$ mod 3 of their position (x,y) on the 12×6 torus; all 24 checks of a class share reset step, CNOT order (labels $A=x^3+y+y^2\mapsto A_1A_2A_3$, $B=y^3+x+x^2\mapsto B_1B_2B_3$), CNOT steps and measurement step, all modulo 18. A window that passes step 17 continues in the next period. Last column: map from the Fig. 12 order of the class.

class	reset	CNOT order	CNOT steps	meas.	map
X_0	17	$A_1A_2A_3B_3B_1B_2$	0, 1, 2, 8, 9, 15	16	rev s_{56}
X_1	15	$A_1A_2A_3B_1B_3B_2$	16, 3, 4, 5, 10, 13	14	rev
X_2	17	$A_3A_1A_2B_3B_1B_2$	0, 1, 5, 6, 7, 15	16	rev s_{56}
Z_0	17	$B_2B_1B_3A_2A_1A_3$	0, 7, 8, 9, 13, 14	15	rev s_{56}
Z_1	2	$B_2B_3B_1A_3A_2A_1$	3, 4, 9, 10, 11, 17	0	rev $s_{12}s_{56}$
Z_2	16	$B_2B_1B_3A_3A_2A_1$	17, 5, 6, 12, 13, 14	15	rev s_{56}

TABLE II. The $[[90,8,10]]$ circuit, period 16. Code: $\ell=15$, $m=3$, $A=x^9+y+y^2\mapsto A_1A_2A_3$, $B=1+x^2+x^7\mapsto B_1B_2B_3$ [1]. All ancillas are reset at step 0 and measured at step 15; steps 11, 13 and 14 contain no CNOT. Every check of a type uses the same order and steps.

type	CNOT order	CNOT steps
X	$A_2A_1A_3B_2B_3B_1$	1, 3, 5, 6, 8, 10
Z	$B_1B_3B_2A_3A_1A_2$	2, 4, 6, 7, 9, 12

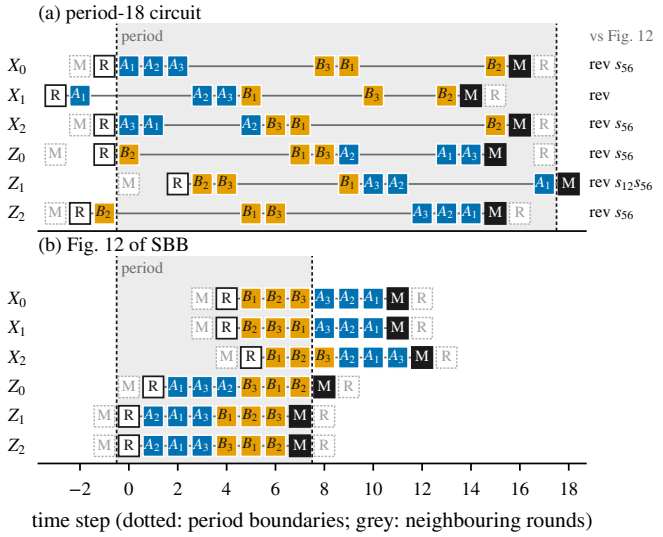


FIG. 2. One period of (a) the circuit of Table I and (b) Fig. 12 of SBB [3], on the same time scale. Rows: the six classes; boxes: CNOTs labelled by the term they implement; R, M: ancilla reset and measurement. Windows crossing the period boundary (dotted) continue into the next period. (b) has period 8 and is not interleaved: on every data qubit the X- and Z-type CNOTs of a round form two contiguous blocks.

tinct columns forming an extended logical. An undetected logical of weight $d_{\rm ext}+1$ consists either of $d_{\rm ext}+1$ faults with distinct nonzero images forming an extended logical, or of $d_{\rm ext}$ such faults forming an extended logical of weight $d_{\rm ext}$ together with one fault of zero image.

Proof. If z faults have zero image and c pairs of nonzero images cancel, the remaining distinct columns form an extended logical of weight $w-z-2c\geq d_{\rm ext}$ by Lemma 1. For $w=d_{\rm ext}$ this forces $z=c=0$; for $w=d_{\rm ext}+1$ it forces $c=0$ and $z\leq 1$. □ Deciding $d_{\rm circ}\geq d_{\rm ext}+2$ is therefore a finite problem: for every extended logical of weight $d_{\rm ext}$ (with or without one zero-image fault) and of weight $d_{\rm ext}+1$, decide whether it lifts.

For the gross circuit, $d_{\rm ext}=11$ and only weight 11 needs to be decided.

Lemma 3 (Finite rounds) *For $R=1,\ldots,32$ and both bases, none of the 4224 weight-11 extended logicals of the gross circuit is the image of eleven undetected faults.*

We verified this by an exact computation: for each logical, a CP-SAT model chooses one DEM mechanism per column subject to every detector sum vanishing; all $2\times 32\times 4224$ instances are infeasible. As a positive control, the same program finds a realisation for all 4224 logicals in Fig. 12. The computation was carried out with two independent implementations. Its inputs, $d_{\rm ext}=11$ and the list of 4224 logicals, are confirmed by two independent exhaustive searches rather than by proof certificates (SM S4, S6). The infeasible lift instances of this finite-R computation were also encoded as CNF and combined into batches by a selector disjunction, so that unsatisfiability of a batch implies that of each member; each batch was refuted with a DRAT proof verified with drat-trim, whose LRAT core was verified with cake_lpr (SM S6).

Lemma 4 (All rounds) *For the gross circuit, if a weight-11 undetected logical existed for some R, one would exist for some $R'\leq 32$.*

Proof sketch. After the last fault of a fault set at layer a, the data frame is constant from layer $a+D-1$ on, $D=18$, because only ancillas still in flight can change it. If an undetected logical had no fault in a window of $G=2D+\Delta-1=52$ layers, where $\Delta=17$ is the longest reset-to-measurement span of a check, no detector could

respond to both the faults before and after the window; both parts would be undetected, their observables would add to a nonzero value, and one part would have weight ≤ 10, contradicting Lemma 1. So a weight-11 logical spans at most 10×51 layers. The noisy circuit is periodic: inserting an 18-layer block at any layer x with $4 \leq x \leq 18(R-1)$ (layers numbered from 0) turns the $(R-1)$-round circuit into the R-round one. Deleting fault-free blocks before and after the logical leaves every detector it touches unchanged, which maps it into a circuit with at most 32 rounds (SM S7). □

Lemmas 1–4 give $d_{\text{circ}} \geq 12$ for every R. For the upper bound, a weight-12 logical operator of the code, applied as single-qubit flips after data initialisation, triggers no detector and flips the logical observable (SM S8). This proves Theorem 1.

Corollary 1 *The Fig. 12 circuit of SBB has $d_{\text{circ}} = 11$ in both bases for every R, as they conjectured.*

This is SBB's Prop. 3 ($d_{\text{circ}} \geq d_{\text{ext}}$) and Cor. 1 ($d_{\text{circ}} \leq d_{\text{ext}}$ for non-interleaved circuits) combined with our value $d_{\text{ext}} = 11$, obtained by a complete symmetric search over the extended code and reproduced with an independently built instance and a second symmetry group. Retiming Fig. 12 without changing its orders cannot reach 12: among schedules of period ≤ 8 there are four conflict-free ones up to translation, all with weight-11 undetected errors, and in the bulk, every legal circuit keeping the orders has such an error in each longer-period family of Table S3 (SM S11).

Gap of two for $[[90, 8, 10]]$. For this code no choice of CNOT orders reaches the code distance through the extended code: for every assignment of per-check orders, the extended code of the X-check side contains a logical of weight at most 8. We verified this with a SAT encoding over all admissible order types whose unsatisfiability is certified by a DRAT proof checked with drat-trim (SM S9). Hence $d_{\text{ext}} \leq 8 = d - 2$ for every single-ancilla circuit of this code, every non-interleaved circuit has $d_{\text{circ}} \leq 8$ by SBB's Cor. 1, and a non-interleaved circuit with the same extended code as Table II attains 8.

Theorem 2 *Under SD6 noise, the circuit of Table II has $d_{\text{circ}} = 10$ in both memory bases for every number of rounds $R \geq 1$.*

Proof. The extended code of its orders has $d_{\text{ext}} = 8$, with exactly 585 logicals of weight 8 and 16830 of weight 9 in each basis (two independent enumerations, a depth-first search and a meet-in-the-middle search). By Lemma 1 no undetected logical has weight ≤ 7. By Lemma 2, one of weight 8 lifts a weight-8 extended logical, and one of weight 9 lifts a weight-9 extended logical or a weight-8 one together with one zero-image fault; in this circuit the zero-image faults that trigger detectors are exactly the ancilla readout flips. For $R = 1, \ldots, 12$ and both bases none of these lifts, as found by three independent implementations. All ancillas are reset at step 0 and measured at step 15, so rounds are separated in time: a minimal undetected logical with a fault-free round inside its span would split into two undetected parts, one of them lighter, so a weight-9 logical spans at most 9 rounds, and deleting fault-free rounds maps it into a circuit with at most 12 rounds. A weight-10 data logical injected after data initialisation gives the upper bound. □

As positive controls, the same lift finds 315 weight-9 lifts for a nearby interleaved circuit with $d_{\text{circ}} = 9$, and 585 weight-8 lifts for a non-interleaved circuit with the same extended code. To our knowledge, the two circuits of this work are the first explicit single-ancilla circuits whose circuit distance exceeds the extended-code distance of every choice of CNOT orders.

Direct cross-check. Both lower bounds were also checked without the extended code, by an exact branch-and-bound search for the lightest undetected logical on the full detector error model, driven by unsatisfied detectors (the same family of methods as the search of SM S4, but an independent implementation with different pruning; its $R = 1$ results were reproduced by a second, independently written implementation). For the gross circuit there is no undetected logical of weight ≤ 11 at $R = 1$, 2 and 3 in both bases, and the weight-12 injection gives the minimum 12. For the $[[90, 8, 10]]$ circuit there is none of weight ≤ 9 for $R = 1, \ldots, 12$ in both bases; with the round-splitting argument above this proves Theorem 2 for every R without any extended-code input (SM S10).

Optimality of the gross period. Within its order class the circuit is as short as a colour-symmetric schedule can be in the bulk frame model described below. Among schedules symmetric under the 24 colour-preserving translations with orders in the class of Fig. 12, every legal schedule with period $P \leq 17$ has a model-liftable weight-11 logical, and so does every schedule with $P \leq 20$ in which all ancillas share one reset and measurement frame; the minimal periods are therefore 18 and 21 in a bulk Pauli-frame model of lifting (SM S11, S12). In this model a data error is seen by a check at the first readout after it, and a logical is model-liftable if its errors can be timed so that every check sees an even number of them between consecutive readouts; a model lift is a genuine weight-11 undetected error for logicals with one length-2 residual (a one-dimensional Helly argument) and was confirmed on the exact detector error model in all 991 sampled cases with several residual columns. The class of orders is itself forced: among all 24 colourings by cosets of translation subgroups of index 2, 3, 4, 6, 8 or 9, with one order type per colour, only the class of Fig. 12 has $d_{\text{ext}} \geq 11$ in both bases. For less symmetric schedules the minimal period is open; in the same model, invariance under a translation subgroup of order 12 already excludes $P \leq 11$, and any non-trivial subgroup excludes

$P = 8$.

Discussion. The obstruction found by SBB is specific to non-interleaved circuits: both the gross code and $[[90, 8, 10]]$ admit single-ancilla circuits that preserve their full distance, and for $[[90, 8, 10]]$ interleaving gains two units over every non-interleaved circuit. The method behind these results is general. Fixing the extended code and searching over orders and timing turns an open-ended circuit search into a lift problem against finite lists of extended logicals of weight d_{ext} and $d_{\text{ext}}+1$ (Lemma 2), which is decided exactly; it applies to every single-ancilla circuit whose distance is limited by Cor. 1 of SBB.

Cost and outlook. The gross circuit uses the qubits of the IBM and SBB circuits, one ancilla per check, but 18 time steps per round instead of 8, with 3168 idle locations per round instead of 288 and 2.8 times the total fault probability per round. We compared it with Fig. 12 and the IBM circuit under the same SD6 noise (idle depolarization on every idle qubit in every layer), $R = 12$, and identical BP-OSD and Relay-BP decoders with exact priors. With full idle noise it is worse at every sampled $p \geq 10^{-3}$: about 1.5×10^3 times worse than the IBM circuit at $p = 2 \times 10^{-3}$ and at least 3 times worse at 10^{-3}, while at matched fault mass the circuits perform alike. Rescaling only the idle channel, $p_{\text{idle}} = \lambda p$, locates the regime where the structure pays off: at $\lambda = 0$ the period-18 circuit has a lower per-round logical error rate than Fig. 12 at $p = 4 \times 10^{-3}$ (ratio 0.37 [0.28, 0.49] with BP-OSD; 0.25 [0.11, 0.60] with Relay-BP) and is statistically indistinguishable from the IBM circuit at $p = 3$–4×10^{-3}, whereas already at $\lambda = 0.1$ it is worse than Fig. 12 by a factor 2.3–3 per round (SM S13). The $[[90, 8, 10]]$ circuit has period 16, three steps of which contain no CNOT, against 8 time steps per round for the depth-7 IBM circuit; its logical error rate has not been evaluated. Shorter periods are therefore the natural next goal: the main open questions are whether $d_{\text{circ}} = 12$ for the gross code and $d_{\text{circ}} = 10$ for $[[90, 8, 10]]$ can be reached at period 8 by less symmetric schedules. Correlated faults, leakage and CZ-native compilations are outside the scope of this work.

Data availability. The circuit specifications, the lists of extended logicals used in the lift computations (4224 of weight 11 for the gross circuit; 585 of weight 8 and 16830 of weight 9 for the $[[90, 8, 10]]$ circuit, each basis), the witnesses, the certificates (clause sets of the unsatisfiable instances) and the reproduction scripts are available at [Zenodo DOI].

AI use. This research was carried out with AI agents (Claude, Anthropic). The agents proposed hypotheses, wrote and ran the search and verification code, worked out the constructions and proofs, and independently rechecked each claim reported here, using separately written code for every computation. The author directed the project, made all decisions at its decision points, and takes full responsibility for the content. In line with APS policy, no AI system is listed as an author.

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Supplemental Material

The sections follow the order of the main text. S1–S2 fix notation and the gross circuits; S3–S8 contain the proof of Theorem 1 and the general lemmas; S9 treats the $[[90, 8, 10]]$ code and Theorem 2; S10 the direct search on the detector error model; S11 Fig. 12 of SBB; S12 the search and the optimality statements; S13 the sensitivity of the logical error rate to idle noise; S14 the verification and the data.

S1. NOTATION AND NOISE MODEL

Code. The gross code [1] has $\ell = 12$, $m = 6$, $A = x^3 + y + y^2$, $B = y^3 + x + x^2$, $H_X = [A \,|\, B]$, $H_Z = [B^T \,|\, A^T]$; qubits and checks are indexed by $i = 6a + b$ for the monomial $x^a y^b$, with left and right blocks of 72 qubits each; $n = 144$, $k = 12$, $d = 12$. Term labels: $A_1 = x^3$, $A_2 = y$, $A_3 = y^2$, $B_1 = y^3$, $B_2 = x$, $B_3 = x^2$. The X check at (x, y) acts on left qubit $(x, y) + A_k$ and right qubit $(x, y) + B_k$; the Z check at (x, y) on left qubit $(x, y) - B_k$ and right qubit $(x, y) - A_k$. The colour of a check is $(x + y)$ mod 3; the 24 translations (a, b) with $a + b \equiv 0 \pmod 3$ preserve colours.

Circuit template. Each check has one ancilla. In every round it is reset, applies its six CNOTs in its order (ancilla as control for X checks, as target for Z checks) and is measured. A schedule assigns each check a reset step, six CNOT steps and a measurement step modulo the period P; windows may wrap around the period boundary, so that rounds of different checks overlap. A Z-basis memory initialises the data in $|0\rangle$, runs R rounds and reads all data in the Z basis; detectors are the first measurement of each Z check, the comparison of consecutive measurements of each Z check, and the comparison of the last measurement with the parity of the final readout. The X basis is analogous.

Noise and d_{circ}. SD6 [15]: after every CNOT a two-qubit depolarizing channel; after every reset a flip; before every measurement a flip; on every qubit idle in a time step a single-qubit depolarizing channel (also during reset and measurement layers). A fault is one Pauli component of one channel. Faults with identical detector and observable signatures are one DEM mechanism, and d_{circ} counts mechanisms. For lower bounds we use the CSS relaxation: in a Z-basis memory only Z-check detectors and the X part of each fault are kept; this can only lower the minimum. Upper bounds are checked on the full circuit. Lower bounds hold for any sub-model of SD6.

S2. THE CIRCUITS

Table I specifies circuit A completely: each of the 144 checks takes the reset step, order, CNOT steps and measurement step of its class. Its resets occur at steps $\{2, 15, 16, 17\}$ and its measurements at $\{0, 14, 15, 16\}$. We verified for circuits A, B and C that every check applies exactly six CNOTs to its support at distinct steps inside its window, that no data qubit takes part in two CNOTs at the same step, that the noiseless circuit fires no detector, and that all detectors are deterministic, for several R in both bases. Interleaving: in circuit A, over one period, 96 data qubits switch four times and 48 twice (cyclically) between X- and Z-type CNOTs.

Circuit B (Table S1) also has period 18; 24 of its checks keep the Fig. 12 order. Circuit C (Table S2) has period 21 and a common frame: all ancillas are reset at step 0 and measured at step 20, so rounds are separated in time. Both have $d_{\mathrm{circ}} = 12$ for every R in both bases (S6, S7).

For reference, the Fig. 12 orders in the same labelling are X_0: $B_1B_2B_3A_3A_2A_1$, X_1: $B_2B_3B_1A_3A_2A_1$, X_2: $B_1B_2B_3A_2A_1A_3$, Z_0: $A_1A_3A_2B_3B_1B_2$, Z_1: $A_2A_1A_3B_1B_2B_3$, Z_2: $A_2A_1A_3B_3B_1B_2$ (period 8, pipelined, non-interleaved).

TABLE S1. Circuit B (period 18). Columns as in Table I.

class	reset	CNOT order	CNOT steps	meas.	map
X_0	0	$B_2B_1B_3A_3A_2A_1$	1, 7, 8, 14, 15, 16	17	s_{12}
X_1	2	$B_2B_3B_1A_3A_2A_1$	3, 6, 11, 12, 13, 0	1	id
X_2	0	$B_2B_1B_3A_2A_1A_3$	1, 9, 10, 11, 15, 16	17	s_{12}
Z_0	1	$A_3A_1A_2B_3B_1B_2$	2, 3, 7, 8, 9, 17	0	s_{12}
Z_1	17	$A_1A_2A_3B_1B_3B_2$	0, 5, 6, 7, 12, 15	16	$s_{12}s_{56}$
Z_2	1	$A_1A_2A_3B_3B_1B_2$	2, 3, 4, 10, 11, 17	0	s_{12}

TABLE S2. Circuit C (period 21, common frame: reset at 0, measurement at 20).

class	CNOT order	CNOT steps	map
X_0	$B_2B_1B_3A_3A_2A_1$	2, 7, 9, 15, 16, 17	s_{12}
X_1	$B_2B_3B_1A_3A_2A_1$	1, 8, 12, 13, 14, 17	id
X_2	$B_2B_1B_3A_2A_1A_3$	1, 10, 11, 12, 16, 18	s_{12}
Z_0	$A_3A_1A_2B_3B_1B_2$	3, 4, 8, 9, 10, 17	s_{12}
Z_1	$A_1A_2A_3B_1B_3B_2$	1, 6, 7, 8, 13, 14	$s_{12}s_{56}$
Z_2	$A_1A_2A_3B_3B_1B_2$	2, 4, 5, 11, 12, 19	s_{12}

S3. INVARIANCE OF THE EXTENDED CODE

Consider a Z-basis memory; only X errors matter. An X error on the ancilla of an X check after its k-th CNOT propagates to the suffix $\{q_{k+1}, \dots, q_6\}$; Z-check CNOTs (data controls, ancilla target) do not change the X frame on the data. Each fault's final data X frame is therefore zero, a single data qubit, or a proper suffix of an X-check order, and its column is $(H_Z s, L_Z s)$. Multiplying by the check itself, which has zero column, a suffix of length 5 is equivalent to $\{q_1\}$ (a data column), of length 4 to $\{q_1, q_2\}$, of length 3 to the complement of $\{q_1, q_2, q_3\}$, and a suffix of length 2 is $\{q_5, q_6\}$. Hence the column set is determined by the three sets $\{q_1, q_2\}$, $\{q_1, q_2, q_3\}$, $\{q_5, q_6\}$ of every check. Reversal maps these to $\{q_5, q_6\}$, the last three and $\{q_1, q_2\}$, which produce the same columns, and s_{12}, s_{56} leave them unchanged. The X basis is symmetric. For circuits A, B and C we also computed the extended codes directly: both bases have 360 merged columns, and the column sets coincide with those of Fig. 12.

S4. THE EXTENDED CODE: $d_{\text{ext}} = 11$ AND THE 4224 WEIGHT-11 LOGICALS

The extended code of Fig. 12 has 360 merged columns per basis (144 of check-weight 3, 144 of weight 6, 72 of weight 9). The 24 colour-preserving translations map each check's order to the order of its image check and preserve the logical classes (checked by solving $L \circ T = ML + NH$ with $\operatorname{rank} M = k$); the columns form 15 orbits.

Search. To decide whether $Hx = 0$, $Lx \neq 0$ has a solution with $|x| \leq W$ we use a complete depth-first search: a minimum-weight solution may be translated so that it contains the representative of its lowest orbit; the search branches on the columns covering an unsatisfied check row, excluding earlier siblings, so that the branches partition the search space; it prunes when the depth plus $\lceil |s|/w_{\max} \rceil$ exceeds W (s the current syndrome, $w_{\max}$ the largest column weight) and closes the last step by table look-up. The program was validated against unpruned meet-in-the-middle enumeration on smaller instances and by checking that split runs add up to unsplit runs.

Results. At $W = 10$ there is no solution in either basis ($\approx 2 \times 10^{10}$ nodes per basis), on two independently constructed instances, and under a second symmetry group of order 8 (45 orbits, $\approx 5 \times 10^{10}$ nodes per basis); weight 11 is attained. Hence $d_{\text{ext}} = 11$. At $W = 11$, two different search programs enumerate the same 4224 extended logicals per basis, all in free orbits of the 24-element group (176 orbits); 360 of them consist of ten data columns and one residual of length 2. Both $d_{\text{ext}} = 11$ and the completeness of the list of 4224 weight-11 logicals rest on two independent exhaustive searches; we did not obtain a checkable proof certificate for them (a direct CNF encoding is beyond CDCL solvers already at weight 6).

S5. PROOF OF LEMMA 1

We work in a Z-basis memory; the X basis follows by exchanging the roles of X and Z and of the two check types. In a Z-basis memory only X components matter. For a fault we track its *data X frame*: the set s of data qubits carrying an X error after the fault has been propagated through the rest of the circuit to the final data readout. Its *image* is the column $(H_Z s,\, L_Z s)$.

Gates of the template. An X-check ancilla is prepared in $|+\rangle$, is the control of its six CNOTs (data qubits as targets) and is measured in the X basis. A Z-check ancilla is prepared in $|0\rangle$, is the target of its six CNOTs (data qubits as controls) and is measured in the Z basis. Under a CNOT, an X on the control is copied to the target, an X on the target stays there, and Z errors never create X errors. Hence: (a) an X on a data qubit never leaves it (as a target it is not copied; as a control it is copied only onto Z-check ancillas, which are never controls and are measured and

reset without acting back on the data); (b) an X on a Z-check ancilla never reaches the data; (c) an X on an X-check ancilla after its k-th CNOT of an instance is copied onto exactly the targets $q_{k+1}, \ldots, q_6$ of the remaining CNOTs of that instance and is then discarded by the next preparation (the X-basis measurement does not see it). These statements use only the direction of each gate, not the relative timing of different checks, so they hold for interleaved and pipelined schedules alike.

Single faults. We go through the SD6 locations. In each case s is empty, a single data qubit, or a suffix $\{q_k, \ldots, q_6\}$ of one X-check order.

1. Data qubits: an X or Y from idle noise, a flip after data preparation, or a flip before the final readout gives $s = \{q\}$ by (a); a Z gives $s = \emptyset$.

2. Z-check ancillas: preparation and measurement flips, idle errors and any error on the ancilla alone give $s = \emptyset$ by (b).

3. X-check ancillas: the preparation flip and the measurement flip are Z errors in this basis and give $s = \emptyset$; an X or Y from idle noise before the first CNOT gives the full support of the check, after the k-th CNOT the suffix $\{q_{k+1}, \ldots, q_6\}$, after the sixth CNOT $s = \emptyset$, by (c).

4. Two-qubit Pauli $P_a \otimes P_d$ after the k-th CNOT of an X check (ancilla a, data q_k): the X part of P_a gives $\{q_{k+1}, \ldots, q_6\}$ and the X part of P_d gives $\{q_k\}$, so $X_a I$ gives $\{q_{k+1}, \ldots, q_6\}$, $I X_d$ gives $\{q_k\}$, and $X_a X_d$ (or any combination with X parts on both) gives their union $\{q_k, q_{k+1}, \ldots, q_6\}$, the suffix starting at q_k.

5. Two-qubit Pauli after a CNOT of a Z check (data q, ancilla): the data part gives $\{q\}$ or $\emptyset$ by (a), the ancilla part gives $\emptyset$ by (b).

Images. A single data qubit gives a data column of the extended code. A suffix $\{q_k, \ldots, q_6\}$ with $2 \le k \le 6$ is a residual, i.e. one of the appended columns (for $k = 6$ it is the data column of q_6). The full support ($k = 1$) is the check itself, whose image is zero because H_Z and L_Z commute with the X stabilisers. So every single fault has image zero or exactly one extended-code column. Merging faults into DEM mechanisms does not change this, since merged faults have identical effects on all detectors and observables, including the augmented ones introduced below.

Telescoping. Augment the circuit with one observable for each row of H_Z and of L_Z, measured on the final data readout. For every Z check i, the sum of all its detectors (first measurement, all consecutive comparisons, final comparison) telescopes to the parity of row i of H_Z on the final readout, i.e. to the augmented observable H_i. Hence an undetected fault set has images whose H parts sum to zero, and whose L parts sum to its logical effect. Its nonzero images therefore sum to an extended-code logical; cancelling equal pairs leaves a set of distinct columns of weight at most the number of faults. If the set has eleven faults and $d_{\text{ext}} = 11$, the eleven images are nonzero, distinct and form one of the listed weight-11 logicals. The template has no X-check detectors in a Z-basis memory; adding them would only remove undetected fault sets, so the bound also holds with all detectors.

Computed check. For every circuit and every R used in S6 we confirmed both facts on the DEM in both bases: each mechanism's per-check detector parity equals the H part of its image, its logical effect equals the L part, and the set of images is exactly the 360 extended columns.

S6. COMPUTATION FOR LEMMA 3

By S5, a weight-11 undetected logical selects, for each of the eleven columns of one listed extended logical, one DEM mechanism with that image, such that the detector sums vanish. For each of the 4224 logicals per basis, a CP-SAT model has a Boolean variable per mechanism of the eleven image groups, an exactly-one constraint per group, and an XOR constraint per detector. Infeasibility for all 4224 logicals at a given R means that no weight-11 undetected logical exists at that R.

Results: circuit A, $R = 1, \ldots, 32$, both bases: all infeasible. A second, independently written lift program agrees and extends the range to $R = 36$. Circuit B: $R = 1, \ldots, 24$; circuit C: $R = 1, \ldots, 14$ and 16, plus a solver-free meet-in-the-middle check (split the eleven groups in two halves, enumerate all detector sums of each half, intersect) for $R \le 3$; all infeasible. Positive control: on Fig. 12 the same programs find a realisation for all 4224 logicals.

Certificates. For circuits A ($R \le 32$), B ($R \le 24$) and C ($R \le 14$ and 16), both bases and all 4224 logicals, the lift instances were encoded as CNF. Instances were combined into batches by a selector disjunction: the batch formula is the disjunction of its members, each guarded by its own selector variable, with no variables shared between members, so that unsatisfiability of a batch implies that of each member. Each batch was refuted by CaDiCaL; each refutation

is a DRAT proof verified with drat-trim, whose LRAT core was verified with the formally verified checker cake_lpr. As a positive control, every one of the 4224 Fig. 12 instances at $R = 1$ is satisfiable and decodes to a valid lift.

S7. PROOF OF LEMMA 4

Notation: period $D = 18$; check q has start s_q, instance j is reset at $s_q + jD$ and measured at $s_q + jD + \Delta_q$ with $\Delta_q \le D - 1$; $\Delta = \max_q \Delta_q = 17$; $S = \max_q s_q - \min_q s_q = 5$. Faults are attached to layers.

(1) Settling. If all faults of a set $F_<$ lie at layers $\le a$, the data frame they produce is constant from layer $a + D - 1$ on: only an ancilla fault can later change the data, through the remaining CNOTs of its own instance, all at layers $\le a + D - 2$; a freshly reset ancilla never pushes errors onto the data (an X-check ancilla only picks up Z errors from the data, a Z-check ancilla only X errors). Every instance reset after $a + D - 1$ is fault-free and measures $H_c e$. If all faults of $F_>$ lie at layers $\ge b$, measurements before b are unaffected.

(2) Splitting. Let $F = F_< \cup F_>$ be a weight-11 undetected logical with no fault strictly between a and b. A comparison detector (c, j) responds to $F_<$ only if $s(c, j-1) < a + D - 1$ and to $F_>$ only if the measurement of instance j, at $s(c, j-1) + D + \Delta_c \le a + 2D + \Delta - 2$, is at a layer $\ge b$. A first-round detector $(c, 0)$ responds to $F_>$ only if $b \le s_c + \Delta_c \le 21$, so with $a \ge 0$ it responds to both parts only if $b - a \le 21 < G$. A final detector responds to both only if $b - a < D + S + \Delta$. Hence for $b - a \ge G = \max(2D + \Delta - 1,\ D + S + \Delta) = 52$ no detector responds to both parts; each part is undetected, their logical effects add to a nonzero value, and one part has weight ≤ 10, contradicting Lemma 1. Consecutive faults of F are therefore less than G layers apart, and F spans at most $10(G-1) = 510$ layers.

(3) Deletion. The noisy SD6 circuit with R rounds equals the circuit with $R - 1$ rounds with an 18-layer block inserted at layer x, for every $x \in [4, 18(R-1)]$ (checked layer by layer in both bases; the template is exactly periodic). Removing a block $[x, x + D)$ that contains no fault of F removes one measurement of every ancilla; a detector of the shorter circuit that does not straddle the cut equals a detector of the longer one, and one that straddles it equals the sum of the two detectors around the removed measurement. F therefore stays undetected, with the same logical effect, provided the frame entering layer $x + D$ of the longer circuit equals the frame entering layer x of the shorter one; this holds if $x + D \le f_1$ (all zero; f_1 the first fault layer) or $x \ge f_{11} + 2D - 2$ (data frame settled by (1), all ancillas in flight fault-free). *(4) Counting.* Write $f_1 \le f_{11}$ for the first and last fault layers of F; by (2), $f_{11} - f_1 \le 10(G - 1) = 510$. *Front:* while $f_1 \ge 4 + D = 22$, the block $[4, 22)$ is fault-free and can be deleted at $x = 4$, shifting F by -18 layers; afterwards $f_1 \le 21$ and hence $f_{11} \le 21 + 510 = 531$. *Back:* a block can be deleted at $x = f_{11} + 2D - 2$ as long as this x is an admissible insertion point, $x \le 18(R-1)$. When no further deletion is possible, $18(R-1) < f_{11} + 2D - 2 \le 531 + 34 = 565$, so $R - 1 \le 31$ and $R \le 32$. (Using $2D - 1$ in place of $2D - 2$, as a more conservative variant does, gives the same bound; the threshold $2D - 3$ would already suffice.) Thus every weight-11 undetected logical of a circuit with R rounds yields one of a circuit with at most 32 rounds, and with Lemma 3 none exists for any R.

For circuits that alternate with their time reverse, Shaw and Terhal reduce any number of rounds to a single round [17]; the gap argument above needs no such symmetry.

Circuits B and C. For circuit C, rounds are separated in time; a fault-free round inside the span of F splits it as in (2), so F occupies at most 11 consecutive rounds, and periodicity reduces any R to $R \le 13$. For circuit B we use a slab version of (2): a fault-free slab from the earliest reset to the latest measurement of one round splits F, provided that on every data qubit every propagating CNOT of round k inside the overlap with round $k + 1$ precedes every reading CNOT of round $k + 1$. We verified this condition for all rounds of the circuits with $R = 3, 8, 24$ in both bases; it reduces any R to $R \le 23$.

S8. UPPER BOUND

For each basis we found a weight-12 logical operator of the gross code of the corresponding type with CP-SAT and applied it as single-qubit flips immediately after data initialisation; each flip is one SD6 mechanism. For $R = 1, 3, 6, 10$ and both bases (and, for circuit A, also at $R = 2$) no detector fires and the logical observable flips; omitting any one of the twelve flips makes the error detectable. Hence $d_{\mathrm{circ}} \le 12$ for circuits A, B and C.

S9. THE [[90, 8, 10]] CODE: BOUND AND THEOREM 2

Code and circuit. $\ell = 15$, $m = 3$, $x = S_{15} \otimes I_3$, $y = I_{15} \otimes S_3$, $A = x^9 + y + y^2$, $B = 1 + x^2 + x^7$, $H_X = [A \,|\, B]$, $H_Z = [B^T \,|\, A^T]$ [1]; $n = 90$, $k = 8$, and an exhaustive search finds no logical of weight ≤ 9 and 558 X-type logicals of weight 10, so $d = 10$. Labels: $A_1 = x^9$, $A_2 = y$, $A_3 = y^2$, $B_1 = 1$, $B_2 = x^2$, $B_3 = x^7$; qubits and checks are indexed by $i = 3a + b$, and the check at (x, y) acts as in S1. The 45 translations are code automorphisms that preserve the labels, and the circuit of Table II is invariant under all of them. It is legal (each data qubit takes part in at most one CNOT per step; every check's six CNOTs cover its support at distinct steps inside its window) and deterministic (each of the 405 overlapping pairs of X and Z checks shares an even number of qubits on which the X CNOT precedes the Z CNOT). X-type CNOTs occupy steps $\{1, 3, 5, 6, 8, 10\}$ and Z-type CNOTs steps $\{2, 4, 6, 7, 9, 12\}$, so each data qubit's cyclic sequence is $XZXZXZ$ and no partition of a round into an X part and a Z part exists.

Every choice of orders has $d_{\text{ext}} \leq 8$. Up to s_{12}, s_{56} and reversal, a check of this code has 90 order types; for 36 of them the extended code of that check alone already has a logical of weight ≤ 8. For the remaining 54, consider each of the 558 weight-10 X-type logicals l: any realised residual columns inside l whose total size reduction is at least 2 (one triple or quadruple, or two disjoint pairs) give, with the rest of l, an extended logical of weight ≤ 8. We encode the choice of one admissible type per check as a SAT instance with one clause for each such combination (17910 combinations, each verified explicitly as an extended logical of weight ≤ 8; 3555 variables and 21690 clauses), in a form that only strengthens the statement (types are required "at least one" per check, and realisation is a one-way implication). CaDiCaL reports unsatisfiability and the DRAT proof is verified by drat-trim. Hence for every assignment of orders the X-side extended code has a logical of weight ≤ 8, and every non-interleaved circuit has $d_{\text{circ}} \leq 8$ by SBB's Cor. 1. A non-interleaved circuit whose extended code equals that of Table II lifts all 585 weight-8 extended logicals at $R = 1$, so 8 is attained.

Lists. The extended code of the orders of Table II has 225 merged columns in each basis (5 orbits under the 45 translations). There are no extended logicals of weight ≤ 7, exactly 585 of weight 8 and 16830 of weight 9 in each basis; a depth-first search and an independent meet-in-the-middle enumeration give identical sets.

Lift. Following Lemma 2, we decide for every weight-8 extended logical whether it lifts, with or without one additional zero-image DEM mechanism, and for every weight-9 extended logical whether it lifts. Zero-image mechanisms with detectors are, in every round, exactly the 45 ancilla readout flips (checked for $R = 1, \ldots, 12$); all zero-image mechanisms of the DEM are admitted as candidates, so this description is not used. For $R = 1, \ldots, 12$ and both bases no instance lifts (full counts, no early stopping), with three implementations: a frontier dynamic program, a CP-SAT model, and a per-logical SAT model; at $R = 1$ an exhaustive enumeration of the products agrees. Controls: a nearby interleaved circuit with $d_{\text{circ}} = 9$ gives 315 weight-9 lifts at $R = 1$ and $R = 2$; the non-interleaved circuit above gives 585 weight-8 lifts, 540 weight-8 lifts with a zero-image fault, and 16830 weight-9 lifts at $R = 1$. The image property (S5) and the telescoping identity were asserted on the DEM for every R used.

All R. In round j all CNOTs lie in layers $16j + 1, \ldots, 16j + 12$ and all measurements in layer $16j + 15$, and the noisy circuit with $R + 1$ rounds equals the R-round circuit with one 16-layer block inserted (checked for $R = 3, \ldots, 14$, both bases). If a minimal undetected logical has a fault-free round strictly inside its span, the faults before that round form an undetected set by themselves (from that round on every measurement reads the same syndrome, consistent with the final readout), so both parts are undetected and one of them is a lighter logical. Hence a weight-9 logical spans at most 9 rounds, and deleting fault-free rounds after the second and before the last maps it into a circuit with at most $2 + 9 + 1 = 12$ rounds.

Upper bound. A weight-10 data logical, applied as flips after data initialisation, fires no detector and flips the observable for $R = 1, 2, 3, 6, 10, 13$ in both bases.

S10. DIRECT SEARCH ON THE DETECTOR ERROR MODEL

The search looks for a minimum-weight set of DEM mechanisms with zero detector sum and nonzero logical effect. It branches on the mechanisms covering an odd detector, excluding earlier siblings so that the branches partition the search space; it prunes any partial set whose detector sum and logical effect both vanish (such a set cannot be contained in a minimal solution), bounds the remaining weight by a packing of detectors with disjoint covering mechanisms, and closes the last level by table look-up. Root cases are split by the orbits of the translations that preserve the detector set and the logical classes (checked by rank conditions). This is the same family of methods as the extended-code search of S4, implemented independently with different pruning; its $R = 1$ results were reproduced by a second, independently written implementation, which also reproduced $d_{\text{ext}} = 11$ and the list of 4224 logicals of S4, and both were tested on 1672 random instances with known answers. The searches produce no proof certificates.

Results. Gross circuit A: at $R = 1$ there is no undetected logical of weight ≤ 11 in either basis and one of weight 12 is found; at $R = 2$ (66 root cases) and $R = 3$ (96 root cases) there is none of weight ≤ 11, and the injection of S8 gives 12. $[[90, 8, 10]]$ circuit: for $R = 1, \ldots, 12$ there is none of weight ≤ 9 in either basis, and weight 10 is found for $R = 1, 2, 3$; the interleaved control circuit gives 9. For the $[[90, 8, 10]]$ circuit, take a minimal undetected logical of weight $w \leq 9$ at any R: by the splitting argument of S9 it spans at most w rounds and maps into a circuit with at most $w + 3 \leq 12$ rounds, contradicting the search. This proves Theorem 2 for every R without extended-code input; it uses the separation and periodicity checks of S9. At $R = 1$ a SAT formulation of the same question on the DEM is also unsatisfiable in both bases.

S11. FIG. 12 OF SBB: $d_{\text{circ}} = 11$ AND RETIMINGS

$d_{\text{circ}} = 11$. Fig. 12 is non-interleaved, so SBB's Prop. 3 and Cor. 1 give $d_{\text{circ}} = d_{\text{ext}}$ for every R; $d_{\text{ext}} = 11$ by S4. The weight-11 extended logicals are realised by eleven circuit faults (checked by injection for $R = 1, 2, 3$ in both bases).

Lift structure. All 4224 weight-11 extended logicals per basis lift to undetected circuit errors of Fig. 12 ($R = 1, 2, 3$).

Retimings with period ≤ 8. Consider periodic schedules with period $P \leq 8$ shared by all checks, one reset, six CNOTs at distinct offsets and one measurement per check and period, and no data qubit in two CNOTs at one step. Weight 6 forces $P = 8$ and all offsets; keeping the Fig. 12 orders leaves one start per check. A backtracking search without symmetry assumptions finds exactly 32 conflict-free assignments, 4 up to global translation (two SAT encodings agree); all four are colour-uniform, deterministic and non-interleaved, and one is Fig. 12 itself. For each of them and every $R = 1, \ldots, 6$ in both bases we injected a weight-11 error that fires no detector and flips the observable.

Longer periods (bulk). For longer periods we use a Pauli-frame model of the bulk. A data X error at time θ on qubit w is first seen by Z check c (readouts at $T_c + nP$) in instance $H_c = \min\{n : T_c + nP > \theta\}$, i.e. for $\theta \in (T_c + (H_c - 1)P, T_c + H_c P)$. A weight-11 logical is *model-liftable* if instance indices H_c can be chosen such that the errors seen by each check can be paired, both errors of a pair being seen in the same instance (so all comparison detectors cancel), and, for every data error, the intervals of its three checks have a common point. For three open intervals of length P on the line, pairwise intersection, which is a difference constraint $H_c - H_d \leq \lfloor (T_d - T_c)/P \rfloor + 1$, implies a common point (one-dimensional Helly); an SD6 fault on w is available there (the two-qubit channel after the CNOT with the latest left endpoint contains $X_w \otimes I$). A residual of length 2 is deposited by an ancilla fault after the fourth CNOT of its check, which fixes its arrival times. *Direction used.* If the difference constraints of a logical of ten data errors and one length-2 residual are feasible, the corresponding eleven faults, placed away from the time boundaries, form a genuine undetected logical error; this was derived independently twice and confirmed by 75 direct injections, each firing no detector and flipping the observable. Non-existence statements need exactly this direction: an unsatisfiable search for a schedule in which no logical is model-liftable shows that every schedule has a model-liftable weight-11 logical, and when the search uses only logicals of the proven form, every schedule has a genuine weight-11 undetected error for sufficiently many rounds. The converse (model blocking implies exact blocking) is not used and fails at the initial time boundary, where the exact lift can succeed via first-round detectors. Blocking a model-liftable logical requires a negative cycle in its constraint graph; we encode this for all modelled logicals, for CP-SAT and in a lazy-clause SAT loop. The certificates block every model-liftable logical. Re-running the same encoders with only the 360 logicals per basis of the form ten data columns plus one length-2 residual gives UNSAT for every row of Table S3, each accompanied by a DRAT proof, so each obstruction rests only on the direction proven above. The resulting status for retimings that keep the Fig. 12 orders is given in Table S3. Per-check-free pipelined schedules with $P \geq 9$ and different CNOT orders are not covered; the latter is how the new circuits escape.

S12. SEARCH AND OPTIMALITY

Search. The circuits were found with SAT models over schedules symmetric under the 24 colour-preserving translations. Variables: for each of the six classes, one of the eight orders of the Fig. 12 class, a start and six CNOT times; constraints: legality (six distinct times modulo P per data qubit), determinism (on every overlapping pair of X and Z checks, the number of shared qubits on which the X instance acts before the Z instance is even, for every instance offset), and, added lazily, one clause per frame-model-liftable weight-11 extended logical forbidding the timing that makes it liftable. Models were then certified with S5–S7. With the Fig. 12 orders pinned the model is unsatisfiable, in line with S9. As a control, 233 random legal deterministic schedules of the common-frame family, sampled without the blocking clauses, all have liftable weight-11 logicals.

TABLE S3. Retimings of Fig. 12 that keep its CNOT orders. "Obstructed": no legal deterministic circuit of the family blocks every model-liftable weight-11 logical. Each row was also re-solved with only the ten-data-plus-one-length-2-residual logicals, which again gives UNSAT; every obstruction is therefore proven.

family	status
24-translation symmetric, common frame, any P	obstructed
24-translation symmetric, pipelined, $P = 8, \ldots, 12$	obstructed
subgroup-invariant pipelined, $P = 9$: three subgroups of order 12, three of order 6, one of order 8, two of order 4 and one of order 3	obstructed
subgroup-invariant pipelined, $P = 10$: one subgroup each of order 12, 8, 6	obstructed
no symmetry, common frame, $P \leq 16$	obstructed (one encoding)
other pipelined families with $P \geq 9$; symmetric pipelined $P \geq 13$	open

Minimum period. With the same model and no circuit fixed, pipelined schedules with $P = 8, \ldots, 17$ and common-frame schedules with $P = 20$ are unsatisfiable. Every unsatisfiable clause set in the data set (129 in total: 104 for the obstruction and optimality instances with all logicals, and 25 length-2-only instances for Table S3) is accompanied by a DRAT proof produced by CaDiCaL and verified with drat-trim; the LRAT core was additionally verified with cake_lpr (SM S14). For the pipelined case a second, independent encoding (absolute times, one-sided difference literals) agrees for $P = 8, \ldots, 17$. The final clause sets were re-solved from scratch by independent SAT solvers (three for $P = 20$ common frame and $P = 12$, two for $P = 17$). Completeness of the family encoding: at $P = 8$ it enumerates exactly the 944 legal deterministic circuits found by an independent backtracking search, and 75 independently sampled legal deterministic circuits at $P = 12$, 17 (pipelined) and 20 (common frame) are all models. Soundness of the clauses: in 991 sampled cases, including logicals with several residual columns, every lift asserted by the frame model was confirmed on the exact DEM, and in 99 sampled cases every violated clause corresponded to an exact lift. Unsatisfiability means that every legal deterministic circuit of the family has a model-liftable weight-11 logical. For logicals of ten data errors and one length-2 residual this implies a genuine weight-11 undetected error for sufficiently many rounds (S9); for logicals with several residual columns the same implication is confirmed in the sampled cases above but not proven in general. The minimum periods are therefore established within the model and refer to sufficiently many rounds. Restricting the CNOT window of class X_0 or of class Z_2 to at most 14 steps is unsatisfiable at $P = 24$, while each of the other four classes can be so restricted; since a window spans at most $P - 3$ steps, this suggests why $P \geq 18$ is needed, but the bound itself rests on the direct runs. At $P = 8$ all 944 legal deterministic circuits of the family were also checked directly: each has a weight-11 undetected error at $R = 1$ in the Z basis.

Uniqueness of the order class. Per check, the extended-code columns are the data columns and three residual sets; up to reversal there are 90 types per basis, of which 45 contain a residual of length 3 that, together with nine data columns, forms a weight-10 extended logical. For each of the 24 colourings by cosets of translation subgroups of index 2, 3, 4, 6, 8 and 9 (3, 4, 3, 12, 1 and 1 subgroups), with one type per colour, a SAT search with witness clauses leaves only the translates of Fig. 12 with $d_{\text{ext}} \geq 11$ in both bases; every witness used was re-verified as at most ten distinct columns of that assignment's extended code with vanishing H part and nonzero L part. Colourings of index 12, 18, 24 and 36 and per-check-free assignments are not covered.

Less symmetry. In the same model, pipelined schedules invariant under the order-12 subgroup generated by $(1, 2)$ are obstructed for every $P \leq 11$, and schedules invariant under any subgroup of order ≥ 2 are obstructed at $P = 8$. Schedules without symmetry, and the remaining subgroups at $P = 12, \ldots, 17$, are open.

S13. SENSITIVITY TO IDLE NOISE

Noise model. Starting from SD6 we multiply only the idle depolarizing channel by λ, $p_{\text{idle}} = \lambda p$; $\lambda = 1$ reproduces SD6 exactly and $\lambda = 0$ coincides with circuit-level depolarizing noise without idle errors (both checked on the detector error models). The CX-native variant uses SI1000 coefficients with CX, RX and MX treated as native operations; it is not the standard SI1000 model. Compiling CX to $H\,\mathrm{CZ}\,H$ and using Z-basis resets and measurements would raise the per-round fault mass of the period-18 circuit by about 12% and that of the period-8 circuits by 6–9%, so the variant slightly favours the period-18 circuit.

Direct sampling. $R = 12$, identical BP-OSD and Relay-BP decoders with exact priors, both bases. Table S4 lists per-round ratios of the logical error rate of the period-18 circuit to Fig. 12 and to the IBM circuit, with 95%

intervals. At $\lambda = 0$ and $\lambda = 0.1$, $p = 4 \times 10^{-3}$, an independent re-sampling with a different noise implementation and BP-OSD code gave 0.37 [0.28, 0.49] and 2.73 [2.14, 3.47] for the ratio to Fig. 12. Several entries rest on few failures; at $p = 2 \times 10^{-3}$ and $\lambda = 0$ no circuit can be distinguished.

TABLE S4. Per-round logical error rate of the period-18 circuit divided by that of Fig. 12 and of the IBM circuit (95% intervals; "–": too few failures).

noise	p	to Fig. 12, BP-OSD	to Fig. 12, Relay-BP	to IBM, BP-OSD	to IBM, Relay-BP
$\lambda = 0$	4×10^{-3}	0.34 [0.19, 0.59]	0.25 [0.11, 0.60]	0.89 [0.45, 1.7]	1.0 [0.33, 2.9]
$\lambda = 0$	3×10^{-3}	0.40 [0.16, 1.02]	0.40 [0.09, 1.9]	1.0 [0.34, 3.0]	0.50 [0.11, 2.4]
$\lambda = 0.1$	4×10^{-3}	3.0 [1.9, 4.7]	2.4 [1.3, 4.5]	3.7 [2.2, 6.0]	7.8 [2.9, 20]
$\lambda = 0.1$	3×10^{-3}	2.5 [0.81, 7.5]	4 [0.56, 22]	3.3 [0.96, 11]	–
$\lambda = 0.1$	2×10^{-3}	2.3 [1.09, 4.95]	–	1.1 [0.19, 4.9]	–
$\lambda = 0.3$	4×10^{-3}	33 [22, 50]	73 [42, 124]	44 [28, 68]	164 [79, 339]
$\lambda = 0.3$	3×10^{-3}	15 [6.9, 33]	20 [7.6, 51]	13 [6.3, 27]	–
CX-native	2×10^{-3}	23 [13, 39]	40 [22, 74]	36 [19, 70]	44 [23, 83]

Per-layer rates. Dividing per-round rates by the period treats every layer as having unit duration, consistent with SD6 but not with physical time. Each round of the period-18 circuit has 12 CX-only layers and 6 layers with measurements or resets; Fig. 12 has 2 and 6. If measurement and reset layers are much longer than CX layers, the ratio of round durations falls from 2.25 towards 1.1–1.3, and the per-layer advantage of the longer period largely disappears.

Leading order. For an ideal decoder at low p, the per-round rate is dominated by weight-12 logical errors with total weight $A_6 p^6$. Summing over the enumerated families (weight-11 logicals and one type of weight-12 logicals), the ratio $A_6(\text{period } 18)/A_6(\text{Fig. } 12)$ is 0.0104 at $\lambda = 0$, 0.084 at $\lambda = 0.1$, 0.84 at $\lambda = 0.3$ and 26 at $\lambda = 1$, crossing 1 near $\lambda \approx 0.32$. Caveats: A_6 is a union-type estimate, not a bound; the other type of weight-12 logical is not enumerated for either circuit, and an unknown factor F in their relative weight moves the crossing (to 0.50 for $F = 0.3$, 0.24 for $F = 2$, 0.20 for $F = 3$); the estimate describes $p \lesssim 10^{-4}$, not the sampled range (at $\lambda = 0.3$ it predicts near equality while direct sampling at 3–4×10^{-3} shows a factor 15–73); and it says nothing about practical decoders.

S14. VERIFICATION AND DATA

All circuits were built and simulated with Stim [16]; exact feasibility questions were decided with CP-SAT and with CDCL SAT solvers, and the extended-code searches use a dedicated depth-first search. The following were obtained with two independent implementations that agree: the construction of the extended code and $d_{\text{ext}} = 11$; the list of 4224 weight-11 logicals; the lift computation of Lemma 3; the enumeration of period-8 schedules; and the minimum-period runs. The arguments of Lemmas 1 and 4 were checked independently, and their structural premises (template identity, image set, periodicity, detector structure) were verified by computer for every circuit. The data set (Data availability) contains the circuit specifications (circuits A, B, C and Fig. 12), the lists of 4224 weight-11 extended logicals for each basis, the witnesses (the weight-12 upper-bound witnesses and the weight-11 witnesses for Fig. 12 and its retimings), the certificates (clause sets of the unsatisfiable instances), and the reproduction scripts for the lower-bound computations; it is available at [Zenodo DOI]. Every unsatisfiable clause set in the data set (129 in total: 104 for the obstruction and optimality instances with all logicals, and 25 length-2-only instances for Table S3), and every batch of the lift computation, is accompanied by a DRAT proof produced by CaDiCaL and verified with drat-trim; the LRAT core was additionally verified with cake_lpr. Here "accompanied" means: proofs of up to 100 MB are included in the data set; larger ones are identified by their SHA-256 hash and can be regenerated byte-identically with the included script. The certificates cover the solver decisions only; the circuit template and the reduction of Lemma 1 are covered by the pen-and-paper proofs and code-level checks of S5. Limitation: $d_{\text{ext}} = 11$ and the completeness of the list of 4224 weight-11 logicals rest on two independent exhaustive searches; we did not obtain a checkable proof certificate for them (a direct CNF encoding is beyond CDCL solvers already at weight 6).